PROGRAMMING FUNDAMENTALS B16
SPRING 2025

Functions

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Outline

- Announcements:
- Please Sign Up on the Circulated Sheet if You want Extra Class on Fri May 02 from 4 to 6 m, If you signup them your attendance will be marked.
- Worksheet One is Uploaded for your Practice, See me During Office Hours for Any Issues Or Email your Availability
- Today
  - Formal Quiz Two
  - Midterm Exam Pattern
  - Revision:
    - File Handling
  - Functions: Parameters By
    - Value
    - Arrays
    - Reference
  - Practice Examples

Formal Quiz 2

Midterm Syllabus and Pattern

PF Midterm Exam Syllabus

File Handling:

- Reading from Text Files
- Reading from CSV Files
- Writing to Text Files
- Writing to CSV Files
- Reading Structure Data from Text or CSV Files
- Writing Structure Data to Text or CSV Files

Functions:

- Passing Parameters by Value, by Reference and by Arrays
- Function Prototypes

PF Midterm Exam Style

Objective:

- Multiple Choice Questions
- Short Question
  - Writing Code Snippets
  - Dry Run
  - Error Finding and Correction
  - Etc.

Subjective:

- Writing Complete C++ Programs, other than main, using more than one functions
- No Restriction on how Parameters are Passed

Midterm Instructions and Guidelines

Midterm Made Easy

Compulsory Instructions (**must follow or lose marks)**

- Follow Good Code Writing Practice:
  - Each variable must be declared in a separate line and must be initialized
  - Use white spaces and indentation in Code to make it Readable
  - Always Write Size of Array in [ ] and Initialize the Array
- Follow Code Writing Template (**given in these slides)**
- Read Question Paper Carefully and Follow Instructions
- Don’t Include Any Libraries other than iostream and fstream
- Don’t use **string** and these built in functions **strlen, getline, get** etc.
- Don’t Use any cin, cout, math, loops or logic in main
- Main must only contain Variables Declarations and Function Calls

Recommendations (**Saves Time and Less Errors)**

- **Take 5 Minutes to Read Paper and Take Notes**
- **Decide and Attempt the Easiest (May be Quickest) Question First**
- **Write Only as Much Code as Required or Asked, e.g.**
  - **Don’t use “include ..” and “using namespace..” for Theory Exam Code**
  - **Don’t Write main function if not Required**
  - **Don’t Repeat Question Description in the Answer Sheet**

Midterm Revision

C++ Code Writing Style Template

Must Follow C++ Code Writing Style Template

// 1. First Line of Code: Code File name e.g. Stats.cpp

// 2. all Includes, for the time being the followings are the only ones to use

```
#include <iostream>

#include <fstream>
```

// 3. Write in comments the description of the program, see example below

```
/*
    Program to calculate sum, average, maximum and minimum values
*/
```

// 4. Prototypes of all Functions

// 5. Function main using Function Prototypes

// 6. Code of all functions

File Reading and Writing

Reading from Text Files

Same as Reading from User:

1. Write Code as Your Are Reading from User
2. Reading from the File is Exactly Like Reading from User
3. File readers works Exactly like cin
4. Make the Changes to make it Read from File
   a) Add **#include <fstream>**
   b) In main add this as the first line**: ifstream reader(“values.txt”);**
   c) Replace cin with reader
   d) Close file after Done Reading: reader.close();

Reading from CSV Files

Same as Reading from User:

1. Write Code as Your Are Reading from a Text File
2. Make these Changes to make it Read from CSV File
    a) **Note that the File Name has now a “.csv”**
    b) **Declare a char Variable e.g. char ch = ‘,’;**
    c) **Read the comma in variable ch and don’t use**

Example:

while(reader >> value) is now while(reader >> value >> ch)

Writing to Text Files

Same as Displaying on Screen

1. Write Code as Your Are Displaying on Screen
2. Writing to File is Exactly Like Displaying on Screen
3. File writers works Exactly like cout
4. Make the Changes to make it Write to File
   a) Add **#include <fstream>**
   b) In main add this line**: ofstream writer(“values.txt”);**
   c) Replace cout with writer
   d) Close file after Done Reading: writer.close();

For Writing to CSB Files: Write a Comma After each value

Example:

writer << value; is followed by writer << ‘,’;

Practice of C++ Code in Functions

C++ Functions

Functions in C++

Why Use Functions in C++?

- Avoids Code Repetition
- A Function can be used many times
- Functions make the Code Modular

What are Function?

- Code Modules to be Used Again and Again
- Each Function has
  - Return Type: a Data Type or void
  - Zero or more Parameters wit each Parameter hs a Data Type
  - Code body that codes the working of the Function

What are Parameters to Functions?

- Arguments that are Passed to the Functions
- Passed **By-Value Parameters** Must Have a Corresponding **Local Variable to Receive a Value**
- Passed **By-Reference Parameters** Must Have a Corresponding **Variable Name for the Memory Address that is Passed. But no Local Variables are Created.**
- **Arrays have Same Value as that of the Address, therefore arrays are always passed by reference**

Practice of C++ Code in Functions

C++ Code in Functions: Practice Learning Process **DON'T USE IN EXAM OR YOU WILL HAVE SHORTGAE OF TIME**

1. Write Code in main only just as you have written in ITC, compile and run
2. Select Portions of code that can be replaced with functions
3. Decide the names of the Functions, Parameters and data type and return type of function
4. Write Function prototypes above main and replace portion of code with call to function one at a time
5. Compile and run
6. Go To Step 2

Example 1

Write Complete C++ Code that reads All Decimal Values from a file "values.txt" in an Array, calculate sum and average, display on screen.

Must use Following Functions:

1. **read:** that reads Decimal Values, one by one, from file "values.txt"
2. **Add:** that adds 2 Decimal values and returns
3. **Average**: that calculates Average rom Sum and Number
4. **display:** that displays Sum and Average Decimals on screen

Example 2

Write Complete C++ Code that reads a word from a user, calculate length, display word and length on screen.

Must use Following Functions:

1. **read:** that reads a Word from a user

2. **length:** that calculates length of a Cstring

3. **display:** that displays a Cstring on screen

Example 3: Simple from File

Write Complete C++ Code that checks a file “prices.txt” containing Prices on separate Lines displays line numbers and corresponding price on Screen. Now Display the Prices in Sorted Descending Order.

Must use Following Functions:

1. **read:** that reads All Decimal Values, in an Array, from file “values.txt”
2. **display:** that displays Line # and Price on screen
3. **sort:** that sorts a double array in Descending Order

Example 4: Moderate from File

Write Complete C++ Code that first read a price value from the user and then Finds if a file “prices.txt” contains this Price. If file contains this price then display the Line #, otherwise display -1 to show that File does not contain the value

Must use Following Functions:

1. **read:** that reads a single decimal value from the user
2. **check**: that checks if a decimal value is present in file “prices.txt” and returns lie number or -1

3. **display:** that displays Line # on screen

Example 5: Moderate from File

Write Complete C++ Code that first reads all decimal values, in an array, from file “values.txt”. Now Checks if values of the array are present in another file “prices.txt” and if for each price that is found display line # and price on screen

Must use Following Functions:

1. **read:** that reads all decimal value from file “values.txt” in an array
2. **check**: that checks if a decimal value is present in file “prices.txt” and returns line number or -1

3. **display:** that displays Line # and a decimal value on screen

Example 6: Moderate from File

A File “product.csv” contains the following data.

```
No., ProductID, Name, Cost,
1, 123,  Pen, 215.25,
2, 234, Marker, 350.5,
3, 345, register, 325.75
```

Another File “sales.csv” contains the following data.

```
No., ProductID, Price,
1,  123, 315.25,
2,  234, 450.5,
3,  345, 425.75
```

Write a C++ Code that calculates profit of each product and saves in a file “profit.csv” in the following format

```
No., ProductID, Profit,
1, 123, 46.46,
2, 234, 28.53,
3, 345, 30.7
```

Example of Code in terms of Functions

Write a complete C++ Program that reads prices from a file, **read path from user**, in an **array** and then displays **Maximum** and **Minimum** prices.

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 3 functions**

Process of Writing Code:

Step 1: Decide names of functions: read, getPath, calculate, display

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Class Activity

Write a complete C++ Program that reads First and Last Names from a user in 2 Cstrings, displays **Each Cstring in all UPPERCASE on screen**.

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads **exactly** 5 words from user. Displays **Each word on a Separate Line on screen.**

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads **a Sentence** from user. Displays **Each word on a Separate Line on screen.**

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads **a para** from a file **para.txt** and Displays **Each word on a Separate Line on screen.**

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads a Paragraph from a File in a Cstring, path read from user. Displays **Each Word and its Length on a Separate Line on screen.**

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads a Paragraph from a File in a Cstring, path read from user. Displays each individual character on screen only once.

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Example

Write a complete C++ Program that reads a Paragraph from a File in a Cstring, path read from user. Write characters and their frequencies to a file, path read from user, as well as displays **on screen (see sample output screen)**

Character: a t h e k
Frequency 2 1 3 5 1

Constraints:

- All Functions, other than main, must have **void** return type
- Main **must not** contain any **cin, cout, file reading or math**
- **Use at least 4 functions**

Show the following Steps of Process of Writing Code:

Step 1: Decide names of functions:

Step 2: Decide parameters of each function:

Step 3: Write Function Prototypes

Step 4: Write Code of main using Function Prototypes

Step 5: Write codes of all functions

Computer Memory

Computer Memory Types and Usage

Two Main Types:

1. Read Only Memory (ROM): Written Once and Read Many Times
   - Retains Value, Even without Power
   - Contains BIOS: Code to Start the Computer and Load Operating System in RAM
2. Random Access Memory (RAM): Can be Written and Read many Times
   - Wipes Out without Power
   - After Startup of Computer by BIOS, the Operating Runs from RAM

RAM is of Two Types:

3. Code RAM: Contains all Executable Code
4. Data RAM: Contains all Variables (Data)

Data RAM is of 2 Types:

5. Stack or Static Memory: Contains all the Variables and Arrays
   - Much Smaller than Heap
   - Compiler Allocates (Creates), within { } and Deletes (Deallocates), or goes out of scope at the End }
     - Compile Time Memory
6. Heap or Dynamic Memory: Contains all the Dynamic Variables and Dynamic Arrays
   - Much Larger than Stack
   - Allocated (Created) and Deleted (Deallocated) Explicitly in Code Through **Pointers**
     - Run-Time Memory

Concept of a Pointer

Pointer: A Variable to hold an Address of a Variable of A Particular Data Type

```
int main()
{
       int x = 5;
       char word[10] = “”;
     // x has an address allocated during Compile Time
     // Address of x cannot be changed
     // x Cannot be deleted in Code
     // x is Deleted at End Bracket } i.e. Goes out of Scope
     int * xp = &x; // a Pointer variable to hold address of data type int
                             // address of x is put in the pointer variable
      return 0;
}
```